Q.28 Give step by step engine ground testing procedure.

Q.29 Compare the construction features of various types of air inlets.

Q.30 Give the basic assumptions made in power cycle.

Q.31 Explain the purpose and difference of secondary and tertiary air.

Q.32 Classify various compressors. Explain any one of them.

Q.33 Draw T-S diagram for turbo shaft engine with thrust augmentation.

Q.34 Explain the operation of bleed valve.

Q.35 Enlist the factors affecting stage pressure rise of an axial flow compressor.

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Differentiate between the components, characteristics, advantage, disadvantages of turbo jet and turbo fan engines.

Q.37 Discuss the properties, advantages and applications of Titanium, cobalt base alloys and reinforced polymer materials used in aircrafts.

Q.38 Explain the theory behind jet propulsion.



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**5th Sem / AME**

**Sub : Turbo Propeller and Jet Engines-I**

Time : 3Hrs. M.M. : 100

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which type of fuel is commonly used in turbojet engines.

a) Gasoline
b) Diesel fuel
c) Kerosene-based jet fuel
d) Hydrogen

Q.2 Thrust can be increases by increasing fuel flow or exhaust_____.

a) Velocity b) Pressure
c) volume d) All of the above

Q.3 A turboshaft engine primarily drives a _______ instead of producing direct thrust

a) Rotor b) Propeller
c) Wheels d) Fan

Q.4 A turboshaft engine delivers power primarily through a ______ shaft.

a) Reciprocating b) Swivelling
c) Rotating d) Levering

Q.5 Turbine blades are made from high-temperature _____ alloys.

a) Duralumin b) Nickel Based

c) PTFE d) Fibres

Q.6 The Brayton cycle is the air standard cycle for :

a) Steam power plants

b) Gas turbine power plants

c) Nuclear power plants

d) Geothermal power plants

Q.7 Compressor blades are made of ______.

a) Tantalum b) Aluminum alloys

c) Titanium d) Carbon fibre

Q.8 Combustion chamber liners must resist high _____ and corrosion.

a) Velocity b) Pressure

c) Temperature d) Iron content

Q.9 Jet propulsion is based on _____ of mass according to Newton 3rd law.

a) Acceleration b) Conservation

c) Ejection d) Deflection

Q.10 Thermal barrier coatings on turbine blades reduce _____ transfer.

a) Heat b) Velocity

c) Power d) Radiation



## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define thrust augmentation.

Q.12 Name any one modern NDT for inspection of hot section.

Q.13 What is pressure ratio?

Q.14 Name any material used in compressor section.

Q.15 Define after burning.

Q.16 Give two advantages of centrifugal compressor.

Q.17 Give one disadvantage of piston engine.

Q.18 What is the function of boroscope?

Q.19 The function of the nozzle in a jet engine is to_____.

Q.20 Supercharger is used to improve_____.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the Brayton's cycle.

Q.22 Compare piston and turbine engine.

Q.23 Explain the working of reduction gear system.

Q.24 Define by-pass ratio.

Q.25 Explain short term engine preservation.

Q.26 Discuss the need of cooling turbine blades.

Q.27 What do you understand by combustion chambers? Explain types of Combustion chambers.